

Explain programming and python in detail

Programming:-

Definition:- It is a process of designing and writing instructions that a computer can follow to perform specific tasks & solve problems

Purpose of programming:-

To solve real world problems, to automate tasks, to develop software applications

Let us go python:- This is a high level, interpreted and general purpose programming language known for its simplicity and readability

Characteristics are:-

- * It is easy to learn
- * It is dynamically typed
- * It is interpreted language
- * It is a cross platform
- * It is a object oriented language
- * This is a large standard library

Applications of python are:-

- * Web development like [Django, flask]
- * Desktop application
- * Data science and analytics
- * It is a scripting and Automation
- * It is used in game development
- * Artificial learning & Machine also

There are basically two types of comments they are

* single line comment

Ex:- # This is single line comment

* Multi line comment

" " This is multi line comment

2

Important of python in modern software development

- * It is much faster for development due to simple syntax
- * It is used for day to day life for the software companies like Google, Netflix, Instagram
- * It is a strong community support
- * Excellent for emerging technologies like AI, ML and data science
- * It is used in calculator to perform calculations
- * It is used in mobile apps to respond the user actions
- * It is used in also ATM for the public transactions.

Describe data type and operators in python with suitable Example

Built in data type

→ Numeric - int, float, complex

x = 10 O/p int

y = 3.14 float

z = 1+2j complex

→ Set types:- set, frozenset

unique - {1, 2, 3}

→ Boolean type:- bool

is_active = True

O/p :- bool

Type identification using type()

Type() returns the data type of variable's value

Ex:-

x = 10

print(type(x))

O/p:- 'int'

Ex:- name = "John"

print(type(name))

O/p:- 'str'

→ Sequence - list, tuple, complex

name = "John" O/p str

nums = [1, 2, 3] list

point = (10, 20) tuple

→ Mapping Type:- dict

Person: { "name": "Alice", "age": 25 }

O/p:- dict

→ None type:- None

Value = None

Ex:- nums = [1, 2, 3]

print(type(nums))

O/p:- 'list'

Operators in python:-

1. Arithmetic:- $+$, $-$, $*$, $/$, $//$, $\%$, $**$

Ex:-	$5+3$	$5-3$	$5*3$	$5/2$	$5//2$	$5\%2$	$2**3$
	$=8$	$=2$	$=15$	$=2.5$	2	1	8
	addition	subtraction	Multiplication	division	floor division	Modulus	Exponentiation

2. Assignment:- $=$, $+=$, $-$, $*=$, $/=$, $\%=$, $**=$

Ex:- $x=5$, $x+=3$, $x-=3$, $x*=3$, $x/=3$, $x\%=3$, $x**=3$

3. Comparison:- (relational):- True/False

equal to	not equal	greater than	less than	greater than equal to
$=$	$!=$	$>$	$<$	$>=$
$5=3$	$5!=3$	$5>3$	$5<3$	$5>=3$

4. Logical Operators:- Used to combine

and	or	not
true is both are true	True if at least 1 is true	reverse result

Ex:- $x>3$ & $x<10$ $x>3$ or $x<2$ not($x>3$)

5. Bitwise operators:- Operator binary numbers

$\&$	$\wedge$	$\sim$	$<<$	$>>$
AND	XOR	NOT	Left shift	right shift

6. Membership operators:- used to test membership

in - present in sequence	not in - not present
--------------------------	----------------------

7. Identity operators:- used to compare memory location

is - same object	is not - different object
------------------	---------------------------

8. Ternary operators:- if - else

Real world usage of operators:-

- * Used in many purpose like calculating salaries
- * Comparing values in decision-making
- * checking if eligibility
- * And last for the validating user input

Input and Output operations in detail

→ Input:- It is data given to the a program like calculating salaries,

It is a data given to the a program from user `input()`

By default, `input()` reads data as a string

Example:-

```
name = input("Enter name:")  
print("hi", name)
```

Output:-

Enter your name :
hi,

→ Output:- It is a result produced by the program

```
print()
```

We can print variables, strings, small text

Example:- `age = 20`

```
print("Age:", age)
```

Output:-

Age: 25

Now Format Output (string) / specifications

```
name =
```

```
age = 20
```

```
print(f"Name: {name}, Age: {age}")
```

Output:-

is 20

Separators in python (`print()` function):-

→ `sep`:- Defines the separator between values default is a space (' ')

→ `end`:- Defines what to print at the end. Default is a newline (\n)

Example:-

```
print("Sun", "Moon", "star", sep=".")
```

```
print("hello", end="!")
```

```
print("world")
```

Output

Sun.M

Type conversion while taking input:-

→ Input () reads data as a string by default

→ convert it to desired type using functions like int(), float(), str(), etc

* Integer input:-

```
age = int(input("Enter age:"))
```

```
print("Age:", age)
```

Output:- 25

25 (integer)

* Float input:-

```
height = float(input("Enter height:"))
```

```
print("Height:", height)
```

Output:- 5.8

Height = 5.8 (float)

* Multiple input:-

```
a, b = map(int, input("enter two numbers:").split())
```

```
print("sum:", a+b)
```

Output:-

10 20

sum = 30

Control statements and decision making statements.

Meaning:- control statements direct the flow of program execution based on conditions & loops.

Importance:- They enable decision-making, repetition and flow control, making program dynamic & functional.

Types of control statements:-

1. Conditional statements:- (if, if-else, if-elif-else):-

It executes code based on conditions

if

x = 10

if x > 5:

```
print("Greater than 5")
```

Output:- Greater than 5

if else

x = 5

if x < 5:

print ("smaller than 5")

else:

print ("less than 5 equal to 5")

if else else

x = 5

if x > 5:

print ("bigger than 5")

elif x == 5:

print ("equal to 5")

else:

print ("less than 5")

2. Looping statements (for, while):-

It repeats code blocks

for loop

for i in range(3):

print(i)

Output:-

0

1

2

while Loop

i = 0

while i < 3:

print(i)

i += 1

Output:-

0

1

2

3. Jump statements (break, continue, pass)

Alter flow within loops or blocks.

break

for i in range(5):

if i == 3:

break

print(i)

Output:-

0

1

2

continue

for i in range(5):

if i == 2:

continue

print(i)

Output:-

0

1

2

3

4

pass (place holder)

if True:

pass # do nothing

No output

Essay on python programming fundamentals:-

→ Role of programming in problem solving

- * Programming helps in breaking complex problems into smaller, manageable steps
- * It allows automation of repetitive tasks, saving time and reducing error.
- * It enables automation of solution making it a powerful tool for solving real-world problems efficiently
- * Logical thinking and algorithm design are developed through programming
- * It enables automation of solution making it a powerful tool.
- * It is a versatile language whose fundamentals empower programming.

→ python syntax simplicity and readability:-

- * python uses simple and clear syntax that is easy to learn for beginners
 - * It closely resembles the english language, making programs easy to read and easy to understand.
 - * Indentation is used instead of brackets with improved code structure and readability
 - * Fewer lines of code are needed compared to many other programming language.
 - * Its clean structure, use of define blocks, and english like keywords make easy
 - * For example, a simple addition operation is expressed as `a+b`.
- Use of comments for code documentation:-
- * Comments are used to explain the purpose of code and improve understanding
 - * Single line comments are written using the `#` symbol.
 - * Comments make programs easier to maintain and debug
 - * They help other programming (and the programmer themselves)

Understand the logic later

- * Triple Quote ... for multi-line comments
- * It improves maintainability functionality
- * Good comments guide others through code's purpose and learning.
- Data types, operators and input/output operation
- * Python supports various data types such as integers, strings and boolean
- * Operators include arithmetic, relational, logical, and assignment operators
- * Input is taken using the `input()` function
- * Output is displayed using the `print()` function
- * These features allow interaction between the user and the program
- * It also supports lists, dictionaries which store and manipulate information
- Control flow using Decision-Making statements
- * Decision-making statements control the flow of a program based on condition
- * Python uses `if`, `elif` and `else` statements for decision-making
- * These statements allow programs to execute different actions for different conditions
- * Control flow makes program dynamic and capable of handling real-world situation
- * Loops (`for`, `while`) handle repetition, while `break` and `continue` modify loop behavior.

Essay on python programming fundamentals:-

→ Role of programming in problem solving

- * Programming helps in breaking complex problems into smaller, manageable steps

- * It allows automation of repetitive tasks, saving time and reducing errors.

- * It enables automation of solution making it a powerful tool for solving real-world problems efficiently

- * Logical thinking and algorithm design are developed through programming

- * It enables automation of solution making it a powerful tool.

- * It is a versatile language whose fundamentals empower programming.

→ python syntax simplicity and readability:-

- * python uses simple and clear syntax that is easy to learn for beginners

- * It closely resembles the english language, making programs easy to read and easy to understand.

- * Indentation is used instead of brackets with improved code structure and readability

- * Fewer lines of code are needed compared to many other programming languages

- * Its clean structure, use of define blocks, and english like keywords make easy

- * For example, a simple addition operation is expressed as `a+b`.

→ Use of comments for code documentation:-

- * Comments are used to explain the purpose of code and improve understanding

- * Single line comments are written using the `#` symbol.

- * Comments make programs easier to maintain and debug

- * They help other programmers (and the programmer themselves)

Solving real-world problems using python programming

Movie Ticket pricing

```
age = int(input("enter your age :"))
```

```
is 3D = int(input("enter 1 if you are watching a 3D movie else enter 0:"))
```

```
if is 3D = 1:
```

```
    if age < 13:
```

```
        print("Ticket price is ₹ 200")
```

```
    elif age > 13 and age < 59:
```

```
        print("Ticket price with extra ₹ 50 is ₹ 300")
```

```
    elif age < 13:
```

```
        print("Ticket price is ₹ 150")
```

```
    elif 13 > age < 59:
```

```
        print("Ticket price is ₹ 250")
```

```
    else:
```

```
        print("Ticket price is ₹ 200")
```

Output:-

```
enter your age : 25
```

```
enter 1 if you are watching a 3D movie else enter 0: 1
```

```
Ticket price with extra ₹ 50 is ₹ 300
```

College Attendance Rule:-

```
per = int(input("enter attendance:"))
```

```
med = int(input("enter 1 if u have certificate else 0:"))
```

```
if per >= 75:
```

```
    print("allowed for exam..")
```

```
elif per >= 60 and med == 1:
```


print ("allowed for exam..")

else:

print ("not allowed for exam..")

Output:-

enter attendance: 55

enter 1 if u have certificate else 0: 0

not allowed for exam

E-commerce Discount:-

bill = int(input("enter bill amount: "))

is_prime = int(input("enter 1 if u are prime member else 0: "))

if is_prime == 1:

if bill >= 5000:

bill = bill * 0.25

print ("bill after discount:", bill)

elif bill >= 2000 and bill <= 4999:

bill = bill * 0.15

print ("bill after discount:", bill)

else:

print ("bill after discount:", bill)

elif bill >= 5000:

bill = bill * 0.2

print ("bill after discount:", bill)

elif bill >= 2000 and bill <= 4999;

bill = bill * 0.1

print ("bill after discount:", bill)

else:

print ("bill after discount:", bill)

Output:-

enter bill amount : 4500

enter 1 if u are prime member 0:0

bill after discount : 4500

Smartphone Battery warning:-

```
perc = int (input ("enter battery percentage :"))
```

```
is charging = int (input ("enter if you are charging else 0:"))
```

```
if is charging == 1:
```

```
    print ("charging")
```

```
elif perc >= 80:
```

```
    print ("normal")
```

```
else:
```

```
    print ("low")
```

Output:-

Enter battery percentage : 80

enter 1 if you are charging else 0:0

Full

Driving Liscense check:-

```
age = int (input ("enter your age:"))
```

```
is test = int (input ("enter 1 if passed test else 0:"))
```

```
if is test == 1:
```

```
    if age >= 18:
```

```
        print ("eligible")
```

```
else:
```



```
print ("not eligible")  
elif age >= 60 and is test == 0
```

```
    print ("eligible")  
else:  
    print ("not eligible")
```

Output:-

```
Enter your age : 68  
enter 1 if passed test else 0:0  
eligible
```

Online Food delivery:-

```
Order = int (input ("enter Order amount:"))  
is gold = int (input ("enter 1 if you are gold member else 0 ="))  
distance = int (input ("enter distance:"))  
if Order >= 500:  
if distance >= 10:  
    print ("delivery charged")  
else: print ("free delivery")  
elif is gold = 1;  
    if distance >= 10:  
        print ("delivery charged")
```

```
else: print ("delivery charged")
```

Output:- enter Order amount : 500

```
enter 1 if u are gold member else 0:1
```

```
enter distance : 15
```

```
delivery charged
```


Bank loan Approval:-

```
Sal = int (input ("enter salary:"))
```

```
C score = int (input ("enter credit score:"))
```

```
if sal > 30000:
```

```
    if C score > 700:
```

```
        print ("loan approved")
```

```
    elif sal > 50000:
```

```
        print ("Loan approved")
```

```
else:
```

```
    print ("Loan rejected")
```

```
else:
```

```
    print ("loan rejected")
```

Output:- enter salary 7850

enter credit score 500

Loan rejected

Electricity Bill:-

```
Units = int (input ("enter no. of units:"))
```

```
if units <= 100:
```

```
    bill = units * 2
```

```
elif units <= 200:
```

```
    bill = (100 * 2) + (units - 100) * 3
```

```
else:
```

```
    bill = (100 * 2) + (100 * 3) + (units - 200) * 5
```

```
print ("Bills is", bill)
```


Output:-

Enter no. of units : 800

Bill is :- 3500

Student Scholarship

```
marks = int (input ("enter marks:"))
```

```
income = int (input ("enter income:"))
```

```
sparents = int (input ("enter is single parents else 0:"))
```

```
if marks >= 85;
```

```
    if income < 50000;
```

```
        print ("gets scholarship")
```

```
    elif sparent = 1;
```

```
        print ("gets scholarship")
```

```
else:
```

```
    print ("not eligible for scholarship")
```

```
else:
```

```
    print ("not eligible for scholarship")
```

Output:-

enter marks: 89

enter income : 596000

enter 1 is single parent else 0: 1

gets scholarship

Online Exam result:-

```
theory = int(input("enter theory marks:"))
Practicals = int(input("enter practical marks:"))
total = theory + practicals
if theory >= 40 and practical >= 40:
    Print("pass")
elif total >= 100:
    print("Pass")
else:
    Print("fail")
```

Output:-

```
enter theory marks: 86
enter practical marks: 24
Pass
```

Hotel Room pricing

```
nod = int(input("enter no. of days:"))
is weekend = int(input("enter 1 if weekend else 0:"))
normal = 3000
weekend = 4000
bill = 1
if is weekend == 1:
    bill = nod * weekend
else:
    bill = nod * normal
```



```
if no d >= 3:
```

```
dis = bill * 0.15
```

```
fin = bill - dis
```

```
print ("final bill is :", fin)
```

Output:-

```
enter no of days: 9
```

```
enter 1 if weekend else 0: 0
```

```
final day bill is : 22950.0
```

Gaming level Unlock:-

```
score = int(input("enter score:"))
```

```
is premium = int(input("enter 1 if premium else 0:"))
```

```
is cheated = int(input("enter 1 if cheated else 0:"))
```

```
if score >= 100 & is premium = 1:
```

```
    if is cheated == 1:
```

```
        print("access denied")
```

```
    else:
```

```
        print("game unlocks next level")
```

```
else:
```

```
    print("access denied..")
```

Output:-

```
enter score: 12
```

```
enter 1 if premium else 0: 1
```

```
enter 1 if cheated else 0: 0
```


Mobile data usage:-

```
data = int(input("enter data used:"))
is unlimited = int(input("enter 1 for unlimited data else 0:"))
is roaming = int(input("enter 1 for roaming else 0:"))
if data <= 2 and is unlimited == 1:
    if is roaming == 1:
        print("unlimited doesn't work")
    else:
        print("unlimited data")
else:
    print("no unlimited data")
```

Output:-

```
Enter data used : 1
enter 1 for unlimited data else 0: 0
enter 1 for roaming else 0: 1
unlimited doesn't work
```

Movie rating display:-

```
avgrate = float(input("enter rating:"))
is editor choice = int(input("enter 1 if it is editor choice else 0:"))
if is editor choice == 1:
    print("recommed..")
elif avgrate >= 8.5:
    print("excellent.")
elif avgrate >= 6.0 and avgrate <= 8.4:
    print("good")
else:
    print("average")
```

Output:-

```
enter rating = 7.5
enter 1 if it is editor choice else 0: 0
good.
```